MODULE 21 · SECTION 3 - TECHNIQUES TO IMPROVE GEN AI MODEL OUTPUT

Grounding and RAG

Three kinds of data to ground a model in, how retrieval augmented generation works, and Google's ways to build it.

SECTION 3 of 4 EXAM WEIGHT ~20% MODULE 21 of 33

admin@sthithapragna.com

RAG turns a closed-book exam into an **open-book exam**, with the right page already open.

Retrieve, augment, generate

01

Retrieve

Search your sources for passages that match the question.

02

Augment

Add those passages to the prompt.

03

Generate

The model answers from them, with citations.

REMEMBER

The name is the method: retrieval, then augmented generation. The model itself never changes.

Three kinds of data to ground in

KIND OF DATA	EXAMPLES
First-party enterprise data	Your documents, databases, policies, catalogue
Third-party data	Licensed specialist data, such as credit ratings
World data	The public web, through Google Search; places, through Maps

THE TIEBREAKER

Ask who owns the facts the answer needs: you, a specialist provider, or the public world.

Grounding with Google Search

- **Connects** the model to fresh, public information on the web.
- **Fixes the knowledge cutoff** for news, events and public facts.
- **Answers include sources,** so people can check them.
- **Knows nothing** about your private data.
- **Google Maps grounding** adds places, routes and opening details.

> THE SIGNAL
>
> Fresh, public facts the model was never trained on: Grounding with Google Search.

How retrieval finds the right page

- **Documents are split** into chunks, such as paragraphs.
- **Embeddings** turn each chunk into numbers that capture its meaning.
- **Similar meanings** sit close together, like books shelved by topic.
- **A question** is turned into numbers too, then matched by meaning.
- **Ranking** puts the most useful passages first.

> REMEMBER
>
> Retrieval matches meaning, not just keywords. "Bring it back" can find the returns policy.

Ways to build RAG on Google Cloud

OFFERING	BEST FOR
Prebuilt RAG with Agent Search	Fastest, fully managed, little ML skill
RAG Engine	A managed framework, your choice of vector store
RAG APIs	Layout parsing, ranking, grounded generation, check grounding
Vector Search	Full control of the vector database
Grounding with Google Search	Public web facts, no data to prepare

THE TIEBREAKER

More control means more work. Start with prebuilt Agent Search unless a stated need rules it out.

Citations: why grounded answers earn trust

- **Every claim** links to the passage it came from.
- **Users check** an answer in seconds, instead of trusting blindly.
- **Auditors** can trace what the system said, and why.
- **Gaps show up:** if no source supports it, the system can say so.
- **Check grounding** scores how well an answer is supported.

> THE SIGNAL
>
> Need answers people can verify, or an audit trail of sources: grounded generation with citations.

RAG or a long context window?

LONG CONTEXT

Put everything in the prompt

- One or a few long documents
- A one-off analysis
- Simple to set up
- Costly and slow if repeated

RAG

Retrieve just what is needed

- Large, growing collections
- Thousands of questions a day
- Lower cost per question
- Citations, and fresh content

THE TIEBREAKER

A few documents, read once: long context can be enough. A large collection queried constantly: RAG.

RAG is only as good as what it retrieves

- **Stale or duplicate sources** give stale or conflicting answers.
- **Badly split documents** separate a rule from its exceptions.
- **Permissions** must carry through: no HR files for store staff.
- **Freshness:** re-index when sources change.
- **Evaluate** retrieval and answers before launch, and after.

THE TAKEAWAY

When a grounded system gives wrong answers, check the sources before the model. Module 12's three handbooks again.

Wrenfield's questions, and where to ground them

QUESTION	GROUND IN
"How long do I have to return a sofa?"	Returns policy: first-party
"Is the autumn range in stock?"	Live catalogue: first-party, via RAG
"What are this week's safety recalls?"	Google Search: world data
"How risky is this new supplier?"	Licensed credit data: third-party
"Which store is nearest to me?"	Google Maps: world data

THE TAKEAWAY

One assistant can ground in several sources. Each question goes to the source that owns its facts.

Phrases that point to each answer

THE QUESTION SAYS	CHOOSE
"Our own documents and records"	First-party grounding
"Licensed data from a specialist provider"	Third-party grounding
"Recent public events"	Grounding with Google Search
"Fastest managed RAG, little ML skill"	Prebuilt RAG with Agent Search
"Choose our own vector store"	RAG Engine or RAG APIs
"Show where each answer came from"	Citations

ON THE EXAM

RAG questions are mostly about the source. Name whose facts are needed, then pick the tool.

> Wrenfield's HR assistant must answer staff questions about holiday, sick pay and shift swaps. The answers live in Wrenfield's own policy documents.

What kind of data should it be grounded in?

A World data, through Google Search

B Third-party data from a provider

C The model's own training data

D First-party enterprise data

◆ PAUSE AND DECIDE

What kind of data should it be grounded in?

A World data, through Google Search

The public web knows nothing of Wrenfield's policies.

B Third-party data from a provider

No provider holds Wrenfield's HR rules.

C The model's own training data

It never saw these documents.

D First-party enterprise data

Wrenfield's own policies, owned by Wrenfield.

THE SIGNAL

Facts the company owns: first-party enterprise data.

> Wrenfield's delivery assistant must warn customers about bank holidays and major public events that affect deliveries, across the UK and the Netherlands.
>
> This information changes year to year and is public.

Which grounding source fits best?

A Agent Search over Wrenfield's documents

B Grounding with Google Search

C Fine-tune the model each year

D The model's built-in knowledge

◆ PAUSE AND DECIDE

Which grounding source fits best?

A Agent Search over Wrenfield's documents
Wrenfield's documents may be out of date on public dates.

B Grounding with Google Search
Fresh, public facts from the web, with sources.

C Fine-tune the model each year
Slow, costly, and still a snapshot.

D The model's built-in knowledge
Limited by the knowledge cutoff.

THE SIGNAL

Fresh, public facts: Grounding with Google Search.

> A Wrenfield director asks for RAG to be explained in one line.
>
> She wants to know how it stops the model guessing.

Which description of RAG is accurate?

A. Fine-tune the model on documents, then answer

B. Generate an answer, then search for sources that agree

C. Retrieve relevant passages, add them to the prompt, then generate

D. Train a new model on all company documents

◆ PAUSE AND DECIDE

Which description of RAG is accurate?

A Fine-tune the model on documents, then answer

That is tuning; RAG leaves the model unchanged.

B Generate an answer, then search for sources that agree

Backwards: retrieval comes first.

C Retrieve relevant passages, add them to the prompt, then generate

Retrieve, augment, generate.

D Train a new model on all company documents

Training, not retrieval.

THE SIGNAL

Retrieve, augment, generate: the model itself does not change.

> Wrenfield's buyers want an assistant that summarises the financial risk of new suppliers.
>
> The answers must draw on licensed credit ratings and company data from a specialist provider.

What kind of grounding fits best?

A Third-party data from the licensed provider

B Grounding with Google Search

C Wrenfield's own supplier files only

D The model's general knowledge

◆ PAUSE AND DECIDE

What kind of grounding fits best?

A Third-party data from the licensed provider
Licensed specialist data, owned by the provider.

B Grounding with Google Search
Public web pages, not licensed ratings.

C Wrenfield's own supplier files only
Useful, but they lack the credit ratings.

D The model's general knowledge
Unverifiable and likely out of date.

THE SIGNAL

Licensed specialist data: third-party grounding.

> Wrenfield's developers are adding RAG to their own app.
>
> They must use the vector database the company already runs, and control how documents are chunked and indexed.

Which offering fits best?

A Prebuilt RAG with Agent Search

B RAG Engine on Agent Platform

C Google AI Studio

D Gemini Notebook

◆ PAUSE AND DECIDE

Which offering fits best?

A Prebuilt RAG with Agent Search

Fastest, but less control over these choices.

B RAG Engine on Agent Platform

A managed RAG framework with a choice of vector store.

C Google AI Studio

A prototyping tool, not a RAG framework.

D Gemini Notebook

A notebook for people, not a RAG framework.

THE SIGNAL

Control over the vector store and chunking: RAG Engine or the RAG APIs.

> Wrenfield's staff assistant must answer from 12,000 documents that change every day.
>
> It handles thousands of questions a day, must cite sources, and must keep cost per question low.

Which approach fits best?

A Put every document in every prompt

B Fine-tune the model every week

C Use the largest model available

D RAG, retrieving only the passages each question needs

◆ PAUSE AND DECIDE

Which approach fits best?

A Put every document in every prompt

Huge cost per question, and may not fit at all.

B Fine-tune the model every week

Stale within a day, and no citations.

C Use the largest model available

Bigger does not know your documents.

D RAG, retrieving only the passages each question needs

Low cost, fresh content and citations.

THE SIGNAL

Large, changing collection, queried constantly: RAG.

> Wrenfield's compliance team will only approve the policy assistant if they can verify every answer.
>
> They need to see exactly which document each statement came from.

Which feature meets this need?

A Grounded answers with citations to the source passages

B Asking the model to state its confidence

C Lowering the temperature

D Longer, more detailed answers

◆ PAUSE AND DECIDE

Which feature meets this need?

A Grounded answers with citations to the source passages

Each statement links to the passage behind it.

B Asking the model to state its confidence

A stated confidence is not evidence.

C Lowering the temperature

Steadier wording; no sources shown.

D Longer, more detailed answers

More words, still no proof.

THE TAKEAWAY

Answers people can verify: citations.

> Wrenfield's grounded policy assistant still gives conflicting answers about delivery charges.
>
> Investigation shows the index holds three versions of the delivery policy, two of them out of date.

What should the team do first?

A Switch to a larger model

B Retrieve more passages for each question

C Clean the sources: remove the outdated versions and re-index

D Fine-tune the model on the current policy

◆ PAUSE AND DECIDE

What should the team do first?

A Switch to a larger model

Reads the same conflicting passages.

B Retrieve more passages for each question

More passages, more conflict.

C Clean the sources: remove the outdated versions and re-index

Fix what is retrieved; the answers follow.

D Fine-tune the model on the current policy

Bakes in one version, but the index still conflicts.

THE SIGNAL

Grounded but wrong: check the sources before the model.

Module 21 in five lines

- RAG: retrieve, augment, generate. The model reads differently; it does not change.
- Ground in first-party, third-party or world data: whoever owns the facts.
- Prebuilt Agent Search first; RAG Engine or RAG APIs for more control.
- Citations make answers checkable; check grounding scores the support.
- Grounded but wrong? Fix the sources before the model.